

Nikita Sajai

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Education

Cornell University

B.A. in Biological Sciences

Concentration in Computational Biology, Minor in Climate Change

Graduated *summa cum laude*, December 2022

GPA: 4.054

Honors Thesis: *The Mutagenic Effect of Recombination*

Research Experience

Junior Specialist

Bruce Wang Lab, UCSF — May 2023 - Present

- Lead data scientist on single-cell and spatial transcriptomics studies of liver development and disease; developed and maintained custom Linux/R/Python pipelines
- **Spatial Transcriptomics of Human Liver:** Curated marker gene panel for cell-type identification and spatial mapping in healthy and diseased livers. Developed spatial data analysis pipeline.
- **Porphyria (PCT) Mouse Hepatocytes:** Characterized transcriptional changes in UROD-treated hepatocytes. Performed pseudotime analyses to map disease progression. Manuscript in preparation.
- **Wild-Type Liver Physiology:** Analyzed sex- and time-dependent variation in hepatocyte gene expression to establish physiological baselines.
- **Bioinformatics Consulting (UCSF Liver Center):**
 - **Acid Ceramidase Loss Effect on Fibrosis Remodeling:** Identified CCL4-specific transcriptional changes in hepatic stellate cells using snRNA-seq analysis.
 - **Visium HD Spatial Transcriptomics:** Integrated Visium HD and scRNA-seq data to study Kupffer cell reprogramming in wild-type vs *Opa1* knockout livers.

- **Proteomics of Liver Perfusate:** Analyzed bile and perfusate data to identify protein signatures of early allograft dysfunction. Manuscript in preparation (second author).
- **ATAC-seq Analysis in MASLD iPSC-Hepatocytes:** Identified differential regulatory elements distinguishing MASLD versus healthy iPSC-hepatocyte chromatin landscapes.

Research Assistant

Pawlowski Lab, Cornell University — Sept 2021 - April 2023

- **Mutagenic Effect of Recombination:** *Honors thesis*; Evaluated the overlap of indels with crossover sites to understand the mutation generation potential of recombination in maize.
- **Mutant Recombination Landscape Effect on Breeding Efficiency:** *Published in PNAS*; Simulated effects of mutant recombination landscapes in rice breeding in efforts to increase global recombination rates and improve the efficiency of breeding programs
- **Using DNA Sonication as Proxy for Chromatin Openness:** Lead researcher; tested the effectiveness of DNA sonication methods to assess chromatin openness in maize.

Summer Research Intern (REU)

Boyce Thompson Institute (Pawlowski Lab) — Jun 2021 - Aug 2021

- Evaluated crossover hotspot proximity to haplotype blocks in maize using Linux/R.
- Analyzed crossover protein localization using immuno-FISH + super-resolution microscopy

Computational & Bioinformatic Skills

Programming & Scripting:

Python, R, Bash

Custom pipeline development for sc/snRNA-seq, ATAC-seq, spatial transcriptomics, proteomics

Reproducible workflows: Git/GitHub, conda, workflow documentation

Bioinformatics Tools:

Alignment/QC: Bowtie2, STAR, Samtools, Bedtools, Picard, MACS2/3, deepTools

Single-Cell & Spatial: Seurat, Scanpy, CellChat, Harmony, Squidpy, Space Ranger

Epigenomics: MACS3, HOMER, featureCounts, ATAC-seq QC (FRiP, TSS enrichment)

Multi-omics: limma, clusterProfiler, KEGG/GO, LASSO regression

Statistical Analysis & Visualization:

R: tidyverse, ggplot2, Seurat, pheatmap, limma, AlphaSimR

Python: pandas, numpy, matplotlib, scikit-learn, scanpy

Experience with mixed-effects modeling, batch correction, pseudotime/trajectory inference

Systems & HPC:

Unix/Linux; Wynton HPC (SLURM), array jobs, pipeline parallelization

Large-scale data management; cloud workflows using Google Cloud Storage buckets and Terra for workflow execution, data organization, and reproducible analysis environments

Data Formats & Resources:

FASTQ, BAM, BED, GTF/GFF, H5AD, Loom

Use of Ensembl/GENCODE, UCSC Genome Browser, MSigDB, KEGG

Publications

- Epstein R, Sajai N, M. Zelkowski, A. Zhou, K.R. Robbins, & W.P. Pawlowski. (2023). "Exploring impact of recombination landscapes on breeding outcomes." *Proceedings of the National Academy of Sciences* (PNAS). 120 (14) e2205785119, <https://doi.org/10.1073/pnas.2205785119> (2023).
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Presentations & Talks

- **Oral Presentation:** *Overview of Liver Gene Analysis Core* - UCSF Liver Center Research in Progress (RIPS), San Francisco, CA, September 2025
 - **Oral Presentation (Lightning Talk):** *Designing a Healthy Human Liver Gene Panel for 10X Xenium Spatial Transcriptomics* - UCSF Liver Center Annual Symposium, San Francisco, CA, May 2024
 - **Poster Presentation:** *The Mutagenic Effect of Recombination* - Maize Genetics Meeting, St. Louis, MO, May 2023
 - **Poster Presentation:** *The Mutagenic Effect of Recombination* - Cornell Honors Thesis Research Symposium, Ithaca, NY, December 2022
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Teaching Experience

Teaching Assistant, Cornell Dept. of Biological & Environmental Engineering *Perspectives in Climate Change Seminar* (Spring 2021, 50 students)

Other Professional Experience

- **Administrative Assistant, arXiv (2022):** Managed pre-print submissions
 - **Research Intern, US Green Chamber of Commerce (2021):** Sustainability research + curriculum design
 - **Advocacy Intern, Common Energy (2021):** Community solar outreach
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Community Involvement

- **Member, oSTEM:** Professional association for LGBTQ+ people in the STEM community — *October 2025 - Present*
 - **Volunteer, Clinic by the Bay** — *Jan 2024 - Present*
 - **Volunteer, Garden for the Environment (San Francisco)** — *Jan 2025 - Present*
 - **Vice President, Young Researchers Program** — *Sept 2019 - Dec 2022*
 - **Member, Human Ecology Sustainability Team** — *Sept 2021 - May 2022*
 - **Member, OASIS: Asian and Pacific Islander LGBTQ+** — *Sept 2021 - Dec 2022*
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Awards

Howard Milstein Scholarship, Cornell University

Awarded Mar 2020, \$5000 scholarship

Tanner Dean Scholar, Cornell University

Awarded Sept 2019, \$1000 scholarship and up to \$5000 research grant